

SAHARSH BHAVE

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EDUCATION

University of Arizona

Master of Science in Data Science

Coursework: Data Analysis and Visualizations, SQL/NoSQL, Data Mining and Warehousing, Introduction to Machine Learning, Big Data Technologies, Applied Natural Language Processing

Tucson, Arizona

Graduation Date: May 2026

Lakshmi Narain College of Technology

Bachelor of Technology in Information Technology

Relevant Coursework: Database Management Systems, Cloud Computing, Artificial Intelligence, Data Structures and Algorithms, Operating Systems, Object Oriented Programming Methods, Computer Networks

Bhopal, India

Graduation Date: May 2024

SKILLS

- **Programming & Data Science:** Python (Pandas, NumPy, scikit-learn, Seaborn, Matplotlib, Streamlit), R, Java, C, LangChain, RAG
- **Databases:** SQL (MySQL, Workbench, PostgreSQL), NoSQL (MongoDB), Data Warehousing
- **Data Engineering/Visualization:** ETL, Data Preprocessing, Power BI, Excel, Git, Jupyter
- **Cloud & Big Data:** AWS, Apache Spark

EXPERIENCE

Graduate Research Assistant – VSI Lab, University of Arizona

ML Heart Failure Prediction Project (for Syncardia Systems, LLC)

Jun 2025 – Present

- Engineered end-to-end data pipelines for patient medical history records, including missing value imputation, feature engineering, and schema design (relational + tabular), reducing inconsistencies by 30% and enabling efficient storage/retrieval for ML workflows.
- Developed and evaluated classification models (Random Forest, XGBoost, SVM) integrated with SQL-driven analytics, achieving ~78% accuracy while ensuring scalable model deployment for healthcare risk predictions.
- Automated experiment tracking and performance monitoring, optimizing recall-precision trade-offs to reduce false negatives in patient risk predictions, and improving reproducibility through Git-based collaboration.

PROJECTS

Company Workforce and Finance SQL Database

- Designed and implemented a relational SQL database from scratch to model company operations, including employees, salaries, departments, and expenditures, improving data accessibility and consistency.
- Applied database normalization and schema design principles to reduce redundancy by 40% and ensure efficient storage of employee-department relationships.
- Developed optimized SQL queries and joins to generate analytical insights on workforce distribution, salary allocation, and departmental expenditure, enabling faster decision-making.

Analyzing Tucson Crime Trends with Unemployment Rate

- Conducted comprehensive analysis of over 25,000 Tucson crime records from 2018 to 2024, identifying a significant -0.6 correlation between crime and unemployment figures, providing data-driven insights into urban safety trends.
- Built Random Forest and Logistic Regression models with 78% accuracy to predict crime trends across 6 Tucson wards.
- Communicated findings through data visualizations (heatmaps, KDE plots, regression charts) to help stakeholders identify high-risk areas.

Customer Churn Predictor

- Built a logistic regression model on 7,000+ Telco records, achieving 73.5% accuracy, 62% precision, and 58% F1 score in predicting customer churn.
- Performed EDA, imputation, feature scaling, and reduced skewness, boosting recall by 15% over the baseline.
- Deployed model with Streamlit for real-time churn prediction, enabling interactive inputs and 95% faster stakeholder testing and interactive scenario analysis.